

MEMO

WTW158 Calculus

Class Test 2 [Mon]

11-15 March 2024

Time: 15 min

8

Surname:

Initials:

Student number:

Group:

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1.5 marks]

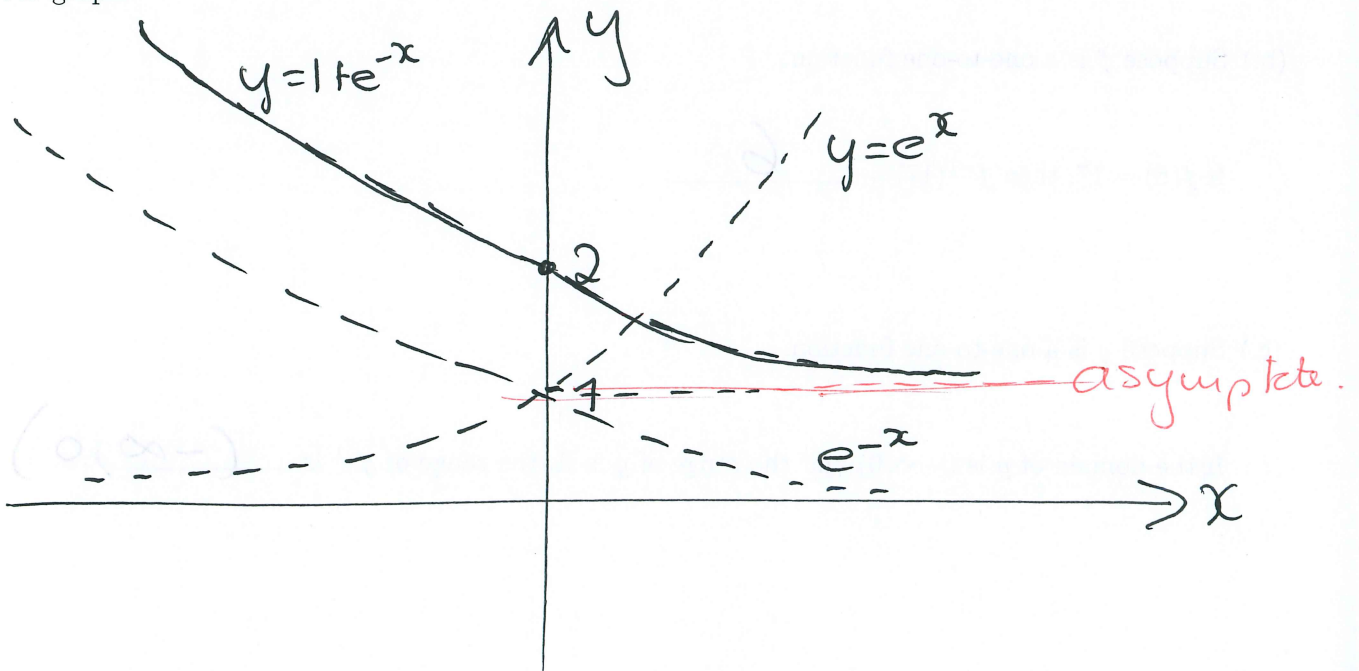
Consider $g(x) = 2x + 1$ and $f(x) = \frac{1}{x}$. Determine the composite function $f \circ g$ and its domain, use interval notation.

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(2x+1) \\ &= \frac{1}{2x+1}, \quad 2x+1 \neq 0 \\ &\quad x \neq -\frac{1}{2} \end{aligned}$$

$$D_{f \circ g} = (-\infty, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$$

QUESTION 2 [2 marks]

Sketch the graph of $y = 1 + e^{-x}$. Show all transformations, as well as all necessary details, e.g., intercepts, on your graph.



QUESTION 3 [2 marks]

Use the laws of exponents to simplify the following expression and leave your answer with positive exponents. Show all your steps.

$$\frac{2x^2}{(3x^{-2}y)^2}$$

$$\begin{aligned} &= \frac{2x^2}{9x^{-4}y^2} \\ &= \frac{2}{9} \frac{x^{2+4}}{y^2} \\ &= \frac{2}{9} \frac{x^6}{y^2} \end{aligned}$$

QUESTION 4 [1.5,0.5,0.5]

(a.) State the horizontal line test (that determines whether a function is one-to-one.)

See p. 54.

(b.) Suppose f is a one-to-one function.

If $f(6) = 17$, then $f^{-1}(17) = \underline{6}$.

(c.) Suppose g is a one-to-one function.

If the domain of g is $(-\infty, 0)$ and the range of g is $\mathbb{R}$, the range of g^{-1} is $\underline{(-\infty, 0)}$.

Memo

WTW158 Calculus

8

Class Test 2 [Tu]

12 March 2024

Time: 15 min

Surname:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Initials:

--	--	--

Student number:

--	--	--	--	--	--	--	--	--	--

Group:

--	--	--

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1.5 marks]

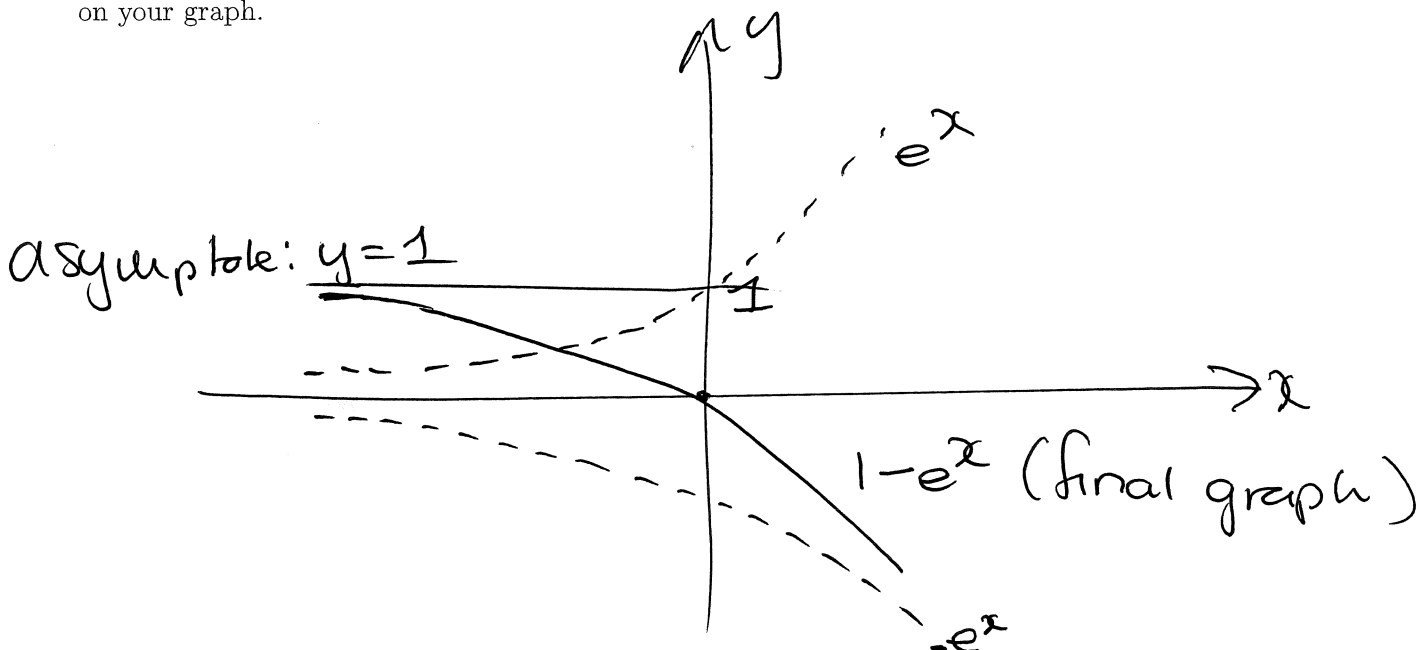
Consider $f(x) = x^2 + 3$. Determine the composite function $f \circ f$ and its domain.

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) \\ &= f(x^2 + 3) \\ &= (x^2 + 3)^2 + 3\end{aligned}$$

$$D_{f \circ f} = \mathbb{R} = (-\infty, \infty)$$

QUESTION 2 [2 marks]

Sketch the graph of $y = 1 - e^x$. Show all transformations, as well as all necessary details, e.g., intercepts, on your graph.



QUESTION 3 [2 marks]

Use the laws of exponents to simplify the following expression and leave your answer with positive exponents. Show all your steps.

$$\frac{\sqrt{a\sqrt{b}}}{\sqrt[3]{ab}}$$

$$= \frac{(a\sqrt{b})^{1/2}}{(ab)^{1/3}}$$

$$= \frac{a^{1/2} (b^{1/2})^{1/2}}{a^{1/3} b^{1/3}}$$

$$= \frac{a^{1/2 - 1/3}}{b^{1/3}} \cdot (b^{1/4})$$

$$= \frac{a^{\frac{3-2}{6}}}{b^{1/3 - 1/4}} = \frac{a^{1/6}}{b^{\frac{4-3}{12}}} = \frac{a^{1/6}}{b^{1/12}}$$

QUESTION 4 [1.5, 0.5, 0.5]

(a.) State the horizontal line test (that determines whether a function is one-to-one).

see p. 57.

(b.) Suppose f is a one-to-one function.

If $f^{-1}(4) = b$, then $f(b) = \underline{4}$.

(c.) Suppose g is a one-to-one function.

If the domain of g is $(0, \infty)$ and the range of g is $\mathbb{R}$, the domain of g^{-1} is $\underline{\mathbb{R}}$.

MEMO

Thursday

WTW158 Calculus

Class Test 2 [2 marks]

11-15 March 2024

Time: 15 min

8

Surname:

Initials:

Student number:

Group:

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks] [2.5 marks]

Consider $f(x) = \sqrt{25 - x^2}$ and $g(x) = \sqrt{x + 1}$. Determine $\frac{f}{g}$ and its domain, use interval notation.

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\sqrt{25 - x^2}}{\sqrt{x + 1}}$$

$$x + 1 > 0 \Rightarrow x > -1$$

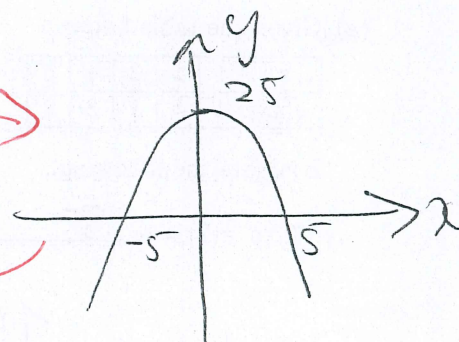
and

$$25 - x^2 \geq 0$$

$$\therefore x \in [-5, 5]$$

$$x \in [-5, 5] \cap (-1, \infty)$$

$$\Rightarrow \underline{x \in (-1, 5]} = D_{f/g}$$



Sketch MB
to solve
inequality

QUESTION 2 [2 marks]

Given the graph of e^x . Write down the equation of the function obtained if the graph of e^x is

- (a) reflected about the y -axis and shifted 3 units upwards

$$\underline{e^{-x} + 3}$$

- (b) shrunk vertically by a factor of 2 and shifted 3 units to the right.

$$\underline{\frac{1}{2} e^{(x-3)}}$$

QUESTION 3 [1.5 marks]

- (a.) Use the laws of exponents to simplify the following expression and leave your answer with positive exponents. Show all your steps.

$$\begin{aligned} & \frac{b^3(3b^{-1})^2}{3\sqrt{b}} \\ &= \frac{b^3 3^2 b^{-2}}{3 b^{1/2}} \\ &= 3 \frac{b^1}{b^{1/2}} = 3 b^{1-1/2} \\ &= \underline{\underline{3 b^{1/2}}} \end{aligned}$$

QUESTION [0.5, 0.5, 0.5, 0.5]

- (a) Given the table below.

x	1	2	3	4	5
$g(x)$	4	5	1	3	2

Evaluate the following.

(i) $g^{-1}(2) = \underline{5}$

(ii) $g^{-1}(g(1)) = \underline{1}$

(iii) $g(g(4)) = \underline{g(3)}$
 $= \underline{1}$

Suppose $g^{-1}(2) = a$
 $\Rightarrow g(a) = 2 \Rightarrow a = 5$

(Suppose $g(1) = 4$
 $g^{-1}(4) = b$
 $\Rightarrow g(b) = 4 \Rightarrow b = 1$)

- (b) Suppose g is a one-to-one function.

If the range of g is $(-1, \infty)$ and the domain of g is $(-\infty, 3)$, then the domain of g^{-1} is $\underline{R_g = (-1, \infty)}$.

MEMO

FRIday.

WTW158 Calculus

Class Test 2 [~~Math~~]

11-15 March 2024

Time: 15 min

8

Surname:

Initials:

Student number:

Group:

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks]

Consider $f(x) = \sqrt{2x+1}$ and $g(x) = x+1$. Determine the composite function $f \circ g$ and its domain, use interval notation.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x+1) \\ &= \sqrt{2(x+1)+1} \\ &= \sqrt{2x+2+1} \\ &= \sqrt{2x+3}\end{aligned}$$

$$\begin{aligned}2x+3 &\geq 0 \\ 2x &\geq -3 \\ x &\geq -\frac{3}{2}\end{aligned}$$

$$D_{f \circ g} = \left[-\frac{3}{2}, \infty\right).$$

QUESTION 2 [2 marks]

Given the graph of e^x . Write down the equation of the function obtained if the graph of e^x is

- (a) reflected about the x -axis and shifted 4 units to the left

$$\underline{-e^{x+4}}$$

- (b) stretched vertically by a factor of 2 and shifted 3 units ^{then} ~~to the right~~ downwards.

$$\underline{2e^x - 3}$$

QUESTION 3 [2 marks]

- (a.) Use the laws of exponents to simplify the following expression, and ~~leave your answer with positive exponents~~. Show all your steps.

$$\frac{2x^3 \cdot x^n}{(2x^{n+1})^2}$$

$$= \frac{2x^{3+n}}{2^2 (x^{n+1})^2}$$

$$= \frac{2x^{3+n}}{4x^{2n+2}}$$

$$= \frac{1}{2} x^{3+n-2n-2} = \frac{1}{2} x^{1-n}$$

QUESTION 4 [0.5, 0.5, 0.5, 0.5]

- (a) Given the table below.

x	1	2	3	4	5
$g(x)$	3	4	5	2	1

Evaluate the following.

(i) $g^{-1}(2) = \underline{4}$

(ii) $g^{-1}(g(1)) = \underline{1}$

(iii) $g(g(4)) = \underline{g(2)} = 4.$

$$\begin{aligned} g^{-1}(2) &= a \\ \Rightarrow g(a) &= 2 \\ \Rightarrow a &= 4. \end{aligned}$$

- (b) Suppose f is a one-to-one function.

If the range of f is $(1, \infty)$ and the domain of f is $(0, \infty)$, then the domain of f^{-1} is $\underline{R_f = (1, \infty)}$